

INTEGRATING MORGAN FINGERPRINTS AND PHYSICOCHEMICAL DESCRIPTORS FOR SOLUBILITY PREDICTION

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INTRODUCTION

Why Solubility Matters?

Aqueous solubility is a critical property in drug discovery because it influences absorption, bioavailability, and formulation. Existing prediction methods rely either on molecular structure representations (fingerprints) or physicochemical descriptors, but each captures only part of the factors governing solubility.

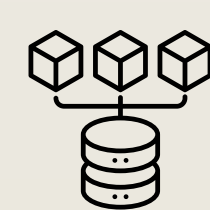
Project Question

Can a hybrid model combining cheminformatics fingerprints and computational chemistry descriptors improve solubility prediction compared with either approach alone?

RESEARCH OBJECTIVES

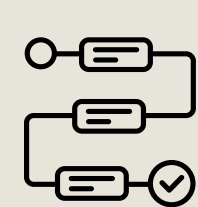
- Compare fingerprint-based and descriptor-based molecular representations and investigate key physicochemical factors affecting solubility.
- Develop fingerprint-only, descriptor-only, and hybrid predictive models, and evaluate their performance using R^2 , RMSE, and cross-validation.
- Assess whether hybrid representations improve accuracy and interpretability, providing insights for drug design and molecular optimization.

METHODOLOGY



Data Collection

- Sources: ESOL (1,128 compounds)¹ + AqSolDB² (9,982 unique compounds) – merged and deduplicated using canonical SMILES and InChIKeys.
- Cleaning: Removed invalid SMILES, standardized structures, filtered outliers ($\log S < -15$ or > 2), kept only measurements at ~ 298 K and pH 7 ± 1 .
- Final dataset: $\sim 9,980$ unique compounds.



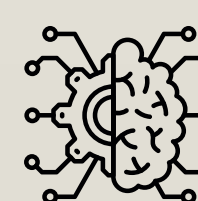
Feature Engineering

Cheminformatics: Morgan fingerprints (radius 2, 2048 bits) using RDKit.

Computational Chemistry: six^{3,4,5} physicochemical descriptors computed with RDKit:

- MolLogP
- TPSA (Topological Polar Surface Area)
- Fsp³ (Fraction of sp³ carbons)
- Number of aromatic rings
- HBD/HBA (hydrogen bond donors/acceptors)

Hybrid: concatenation of fingerprints³ and descriptors (2054 features total).



Model Training & Evaluation

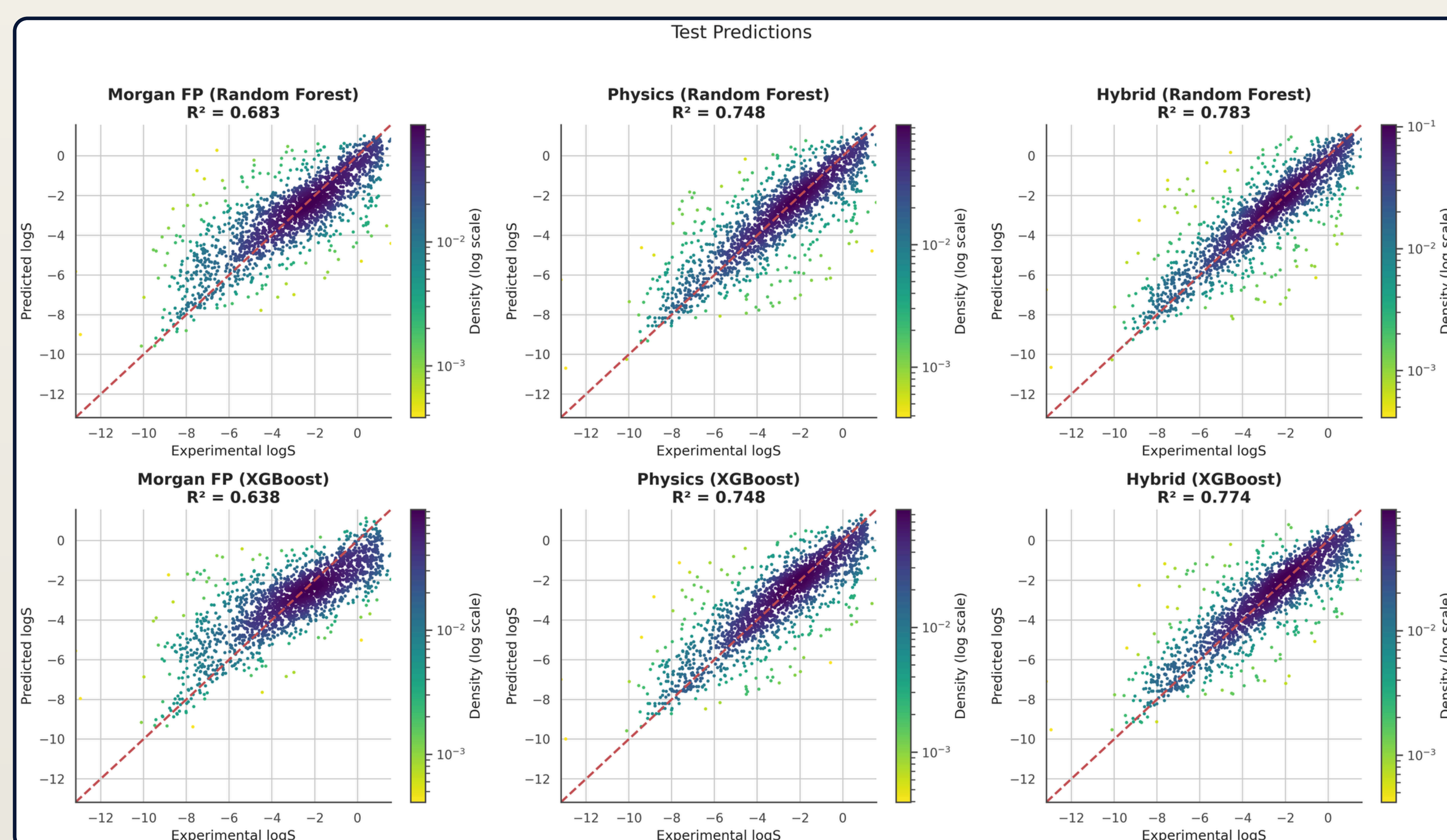
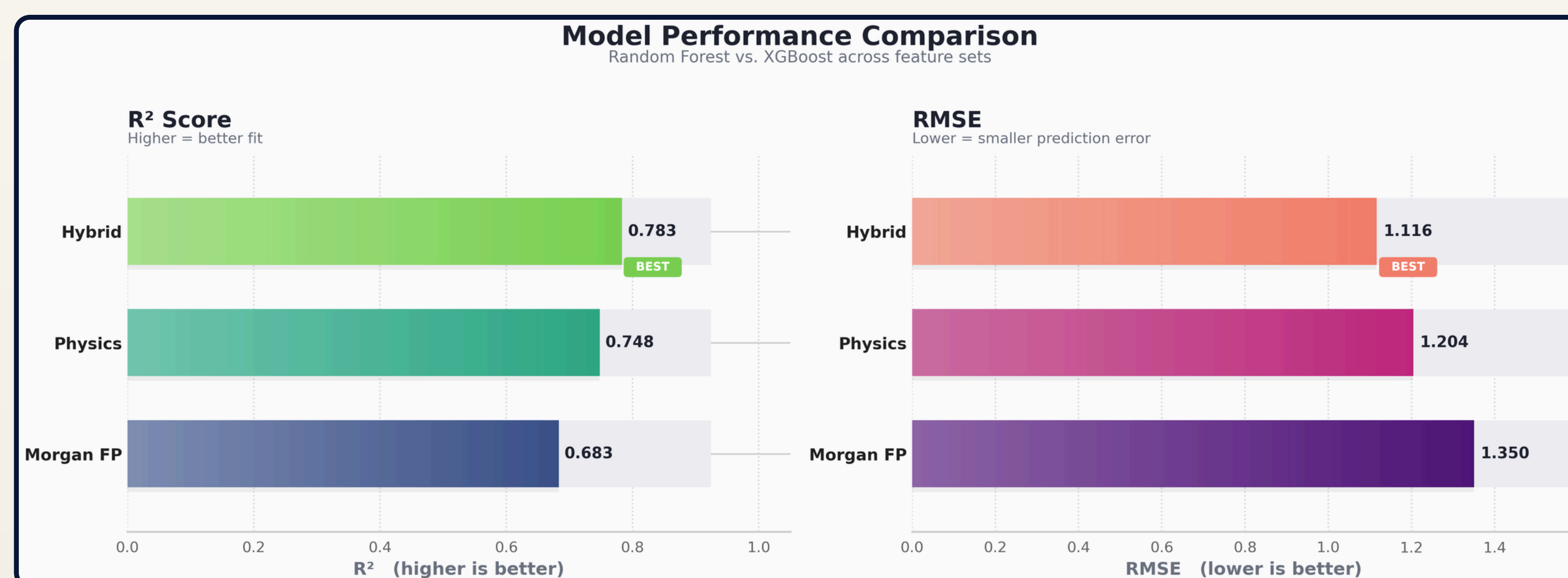
Algorithms: Random Forest (100 trees) and XGBoost (100 trees, learning rate 0.1).

Split: 80% training, 20% test set (stratified random split, fixed seed).

Cross-validation: 5-fold CV on training set for hyperparameter stability.

Metrics: R^2 (coefficient of determination) and RMSE (root mean squared error).

RESULTS AND DISCUSSION



Random Forest and XGBoost models were evaluated on a held-out test set (20% of data). The hybrid model (Morgan fingerprints + physicochemical descriptors) achieved the best predictive performance, yielding the highest R^2 score and outperforming models based solely on cheminformatics features or physicochemical descriptors. MolLogP, TPSA, and Fsp³ are the most influential physicochemical descriptors. The hybrid model consistently improves predictive performance, and the interactive Streamlit web app makes it accessible for practical use in drug discovery.

CONCLUSION

The hybrid representation strategy (fingerprints + descriptors) consistently improves solubility prediction across both Random Forest and XGBoost models. Our hybrid RF model ($R^2 = 0.783$) is competitive with the Morgan FP alone model of Tayyebi & al. ($R^2 = 0.810$)³ while using only 6 descriptors instead of 2048 bits.

Our results demonstrate that combining structural connectivity with physicochemical properties yields a robust, interpretable model suitable for real-world drug discovery workflows. The accompanying Streamlit web app offers an accessible tool for medicinal chemists to rapidly screen compounds before synthesis.

Try the Web App!

Scan the QR code below to predict solubility from any SMILES string:



<https://solubility-predictor-vpm.streamlit.app/>

ACKNOWLEDGMENT

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